

CBSE CLASS 9 MATHEMATICS

Formula & Concept Bible

2025–26 · Official CBSE Syllabus · Extensive Concepts

6 Units · 12 Chapters · 180+ Formulas · All Theorems · Worked Examples · SA Exam Prep

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Official Syllabus Overview 2025-26 (Theory = 80 Marks + 20 Internal)

Unit	Chapters	Marks
I: Number Systems	Ch 1: Real Numbers	10
II: Algebra	Ch 2: Polynomials Ch 3: Linear Equations in Two Variables	20
III: Coordinate Geometry	Ch 4: Coordinate Geometry	04
IV: Geometry	Ch 5: Euclid's Geometry Ch 6: Lines & Angles Ch 7: Triangles Ch 8: Quadrilaterals Ch 9: Circles	27
V: Mensuration	Ch 10: Heron's Formula Ch 11: Surface Areas & Volumes	13
VI: Statistics	Ch 12: Statistics (Bar Graph, Histogram, Frequency Polygon)	06
	TOTAL THEORY	80



How to Use This Book

- Each chapter follows: Concept Explanation → Visual Graphs → Theorems (with proof outline) → Formulas → Common Mistakes → Worked Example → Practice Questions
-  Important = high-probability exam theorems and formulas
-  Mistake boxes = errors that cost marks every year
-  Concept boxes = step-by-step proof methods
-  Theorem boxes = statement + proof strategy
- Visit sjmaths.com for free video solutions and past paper practice

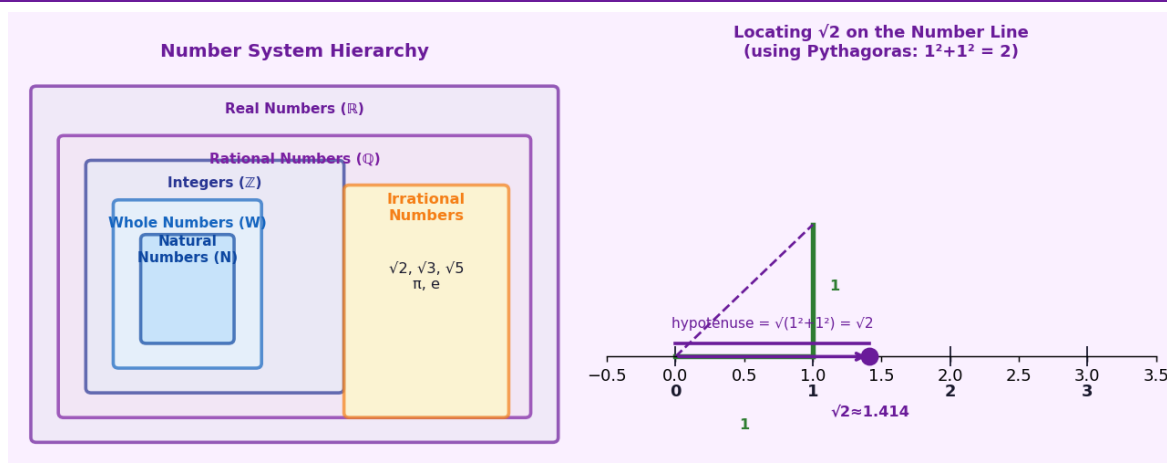


CHAPTER 1 · Unit I — 10 Marks

Real Numbers

Number Systems | Irrational Numbers | Laws of Exponents

1.1 The Number System Hierarchy

Graph 1.1: Number system hierarchy (left) and locating $\sqrt{2}$ on the number line (right)

Number Set	Definition	Examples
Natural (N)	Counting numbers starting from 1	1, 2, 3, 4, 5, ...
Whole (W)	Natural numbers + zero	0, 1, 2, 3, 4, ...
Integer (Z)	Positive, negative and zero	-3, -2, -1, 0, 1, 2, 3, ...
Rational (Q)	Can be written as p/q where $p, q \in \mathbb{Z}$, $q \neq 0$	$3/4$, $-2/5$, 0.75, 0.333...
Irrational	Cannot be written as p/q ; non-terminating, non-recurring	$\sqrt{2}$, $\sqrt{3}$, π , e
Real (R)	All rational and irrational numbers together	Everything on the number line

1.2 Rational Numbers and Decimal Expansions

Key Rules

- Rational number $\rightarrow$ terminating OR recurring decimal expansion
- Irrational number $\rightarrow$ non-terminating AND non-recurring decimal
- Terminating decimal: denominator (in lowest form) = $2^s \times 5^t$ only
- Non-terminating recurring: denominator has prime factors other than 2 and 5

Finding Rational Numbers Between Two Numbers:

- ▶ Between a and b ($a < b$): any number $(a+b)/2$ works | or list decimals between them
- ▶ There are INFINITELY many rationals between any two distinct rationals

1.3 Irrational Numbers

- ▶ $\sqrt{2} = 1.41421356...$ (non-terminating, non-recurring $\rightarrow$ irrational)
- ▶ $\pi = 3.14159265...$ (ratio of circumference to diameter of any circle $\rightarrow$ irrational)
- ▶ Sum / difference / product of two irrationals may be rational: $\sqrt{2} \times \sqrt{2} = 2$ (rational!)
- ▶ Every real number is represented by a UNIQUE point on the number line

🧭 Locating $\sqrt{n}$ on Number Line (Pythagoras method)

- To locate $\sqrt{2}$: draw right triangle with both legs = 1. Hypotenuse = $\sqrt{(1^2+1^2)} = \sqrt{2}$
- To locate $\sqrt{3}$: take hypotenuse $\sqrt{2}$ as one leg, 1 as other leg. New hyp = $\sqrt{(2+1)} = \sqrt{3}$
- To locate $\sqrt{n}$: continue building on previous hypotenuse: $\sqrt{(n-1)}$ and 1 $\rightarrow \sqrt{n}$

1.4 Rationalisation

- ▶ Rationalise $1/(a+b\sqrt{x})$: multiply numerator and denominator by $(a-b\sqrt{x})$
- ▶ $\frac{1}{a+b\sqrt{x}} = \frac{a-b\sqrt{x}}{a^2-b^2x}$ *multiply top and bottom by conjugate $(a-b\sqrt{x})$*
- ▶ Rationalise $1/(\sqrt{x}+\sqrt{y})$: multiply by $(\sqrt{x}-\sqrt{y})$ to get $1/(x-y)$

1.5 Laws of Exponents (Indices)

Law	Formula	Example
Product	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7$
Quotient	$a^m \div a^n = a^{m-n}$	$5^6 \div 5^2 = 5^4$
Power of power	$(a^m)^n = a^{mn}$	$(3^2)^4 = 3^8$
Product rule	$(ab)^n = a^n \times b^n$	$(2 \times 3)^4 = 2^4 \times 3^4$
Quotient rule	$(a/b)^n = a^n/b^n$	$(4/3)^3 = 64/27$
Zero exponent	$a^0 = 1$ ($a \neq 0$)	$7^0 = 1$
Negative exponent	$a^{-n} = 1/a^n$	$2^{-3} = 1/8$
Rational exponent	$a^{(1/n)} = \sqrt[n]{a}$	$4^{(1/2)} = \sqrt{4} = 2$

- ▶ $a^{(m/n)} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$ *Rational exponent: denominator = index of root, numerator = power*

⚠ Common Mistakes

- $a^m \times b^n \neq (ab)^{m+n}$ (different bases cannot be directly added as exponents)
- $(\sqrt{2})^2 = 2$, NOT $\sqrt{4} = 2$. Both give 2, but reason matters in proofs.

- Rationalise by conjugate, not just by multiplying by the denominator.
- $2^{-3} = 1/8$, NOT -8 . Negative exponent means reciprocal, not negative value.

1.6 Worked Example

Worked Example

Q: Simplify: $(3^3 \times 3^{-2}) / (3^4 \times 3^{-3})$

Step 1: Numerator: $3^3 \times 3^{-2} = 3^{(3-2)} = 3^1 = 3$

Step 2: Denominator: $3^4 \times 3^{-3} = 3^{(4-3)} = 3^1 = 3$

Step 3: Result: $3/3 = 1$

∴ Answer: 1

1.7 Practice Questions

Year	SA / LA Question
Type	Represent $\sqrt{5}$ on the number line. Show all construction steps.
Type	Simplify: $(5 + \sqrt{3})(5 - \sqrt{3})$. Is the result rational or irrational?
Type	Rationalise the denominator of $4/(3+\sqrt{5})$.
Type	Simplify: $[(3^2)^3 \times 3^{-4}] / [3^3 \times (3^{-2})^2]$
Type	Prove that $\sqrt{3} + \sqrt{5}$ is irrational.

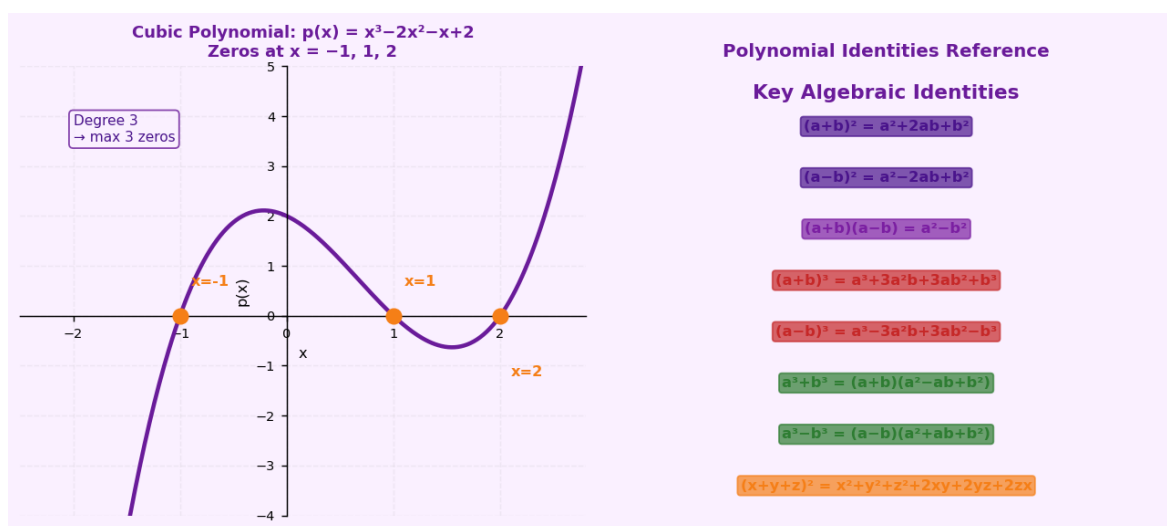
CHAPTER 2 · Unit II — Part of 20 Marks

Polynomials

Zeros | Remainder Theorem | Factor Theorem | Identities

2.1 Key Definitions

Term	Definition	Example
Polynomial	Algebraic expression with non-negative integer powers	$3x^2+2x-5$
Degree	Highest power of the variable	Degree of $5x^3-4x$ is 3
Zero polynomial	All coefficients = 0	0 (degree not defined)
Zero of polynomial	Value of x where $p(x) = 0$	Zero of $x-3$ is $x=3$
Monomial	Polynomial with 1 term	$5x^3$
Binomial	Polynomial with 2 terms	$3x^2-7$
Trinomial	Polynomial with 3 terms	x^2+3x-5



Graph 2.1: Cubic polynomial with 3 zeros (left) and key algebraic identities (right)

2.2 Remainder Theorem



Remainder Theorem

- If $p(x)$ is divided by $(x-a)$, the remainder = $p(a)$
- To find remainder when $p(x)$ is divided by $(x-3)$: just calculate $p(3)$
- Proof: $p(x) = (x-a) \times q(x) + r$. Set $x=a$: $p(a) = 0 \times q(a) + r = r$

- ▶ Remainder when $p(x)$ is divided by $(x-a) = p(a)$
- ▶ Remainder when $p(x)$ is divided by $(ax-b) = p(b/a)$

2.3 Factor Theorem

Factor Theorem (derived from Remainder Theorem)

- $(x-a)$ is a FACTOR of $p(x)$ if and only if $p(a) = 0$
- If $p(a) = 0$, then $(x-a)$ is a factor and divides $p(x)$ exactly (remainder = 0)
- Use to factorise cubic polynomials: guess a root (usually $\pm 1, \pm 2, \pm 3$), verify, then divide

2.4 Algebraic Identities (must memorise ALL)

- ▶ $(a+b)^2 = a^2 + 2ab + b^2$
- ▶ $(a-b)^2 = a^2 - 2ab + b^2$
- ▶ $(a+b)(a-b) = a^2 - b^2$
- ▶ $(x+a)^2 - (x+b)^2 = (a-b)(2x+a+b)$
- ▶ $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
- ▶ $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
- ▶ $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
- ▶ $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- ▶ $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
- ▶ $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Special case of last identity

- If $a+b+c = 0$, then $a^3+b^3+c^3 = 3abc$ (very useful shortcut!)
- Example: if $x+y+z=0$, then $x^3+y^3+z^3 = 3xyz$

Common Mistakes

- $(a+b)^2 = a^2+2ab+b^2$, NOT a^2+b^2 . The middle term $2ab$ is always forgotten.
- $(a+b)^3 \neq a^3+b^3$. Use the full expansion with $3ab(a+b)$.
- Factor theorem: $(x-a)$ is a factor when $p(a)=0$, NOT $p(-a)=0$.

2.5 Worked Example

Worked Example

Q: Factorise: $2x^3 + x^2 - 5x + 2$

Step 1: Try $p(1) = 2+1-5+2 = 0$ ✓ So $(x-1)$ is a factor

Step 2: Divide: $2x^3+x^2-5x+2 \div (x-1) = 2x^2+3x-2$

Step 3: Factorise $2x^2+3x-2 = (2x-1)(x+2)$

Step 4: Complete: $(x-1)(2x-1)(x+2)$

$\therefore$ Answer: $(x-1)(2x-1)(x+2)$

2.6 Practice Questions

Year	SA / LA Question
Type	Find the remainder when $p(x)=x^3-4x^2+x+6$ is divided by $(x-1)$.
Type	Factorise: x^3-3x^2-9x-5 using Factor Theorem.
Type	Verify: $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$
Type	Evaluate 99^3 using $(100-1)^3$ identity.
Type	If $x+y+z=0$, show that $x^3+y^3+z^3=3xyz$.

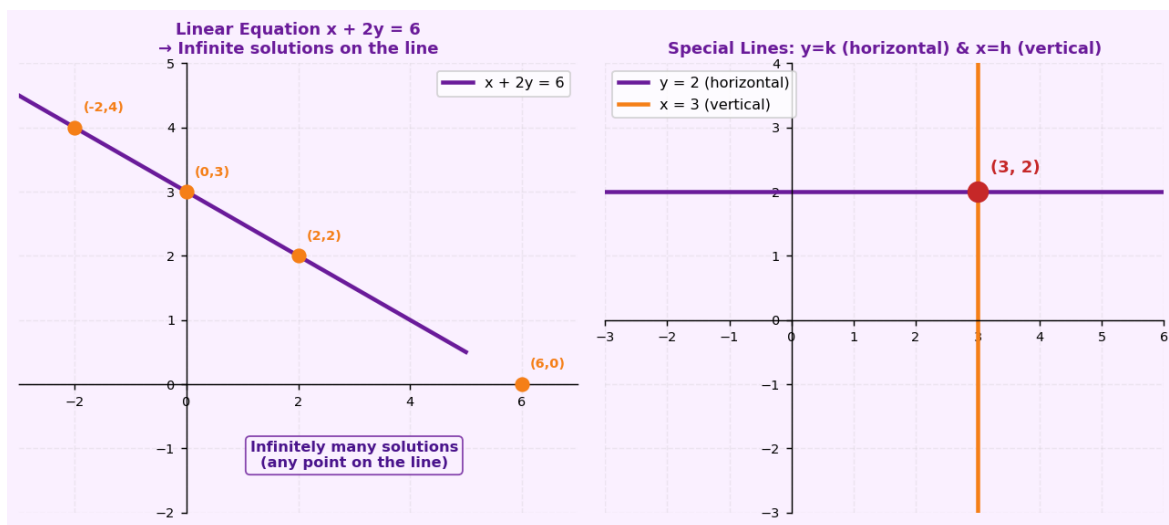
CHAPTER 3 · Unit II — Part of 20 Marks

Linear Equations in Two Variables

 $ax+by+c=0$ | Infinite Solutions | Graphical Representation

3.1 Standard Form and Solutions

- ▶ Standard form: $ax + by + c = 0$, where a, b are not both zero
- ▶ A solution is an ordered pair (x, y) that satisfies the equation
- ▶ Every linear equation in two variables has INFINITELY many solutions

Graph 3.1: Linear equation $x+2y=6$ with multiple solutions (left) and special lines $x=k, y=h$ (right)

3.2 Graphical Representation

- ▶ The graph of a linear equation in two variables is always a STRAIGHT LINE
- ▶ Every point on the line is a solution; every solution lies on the line
- ▶ $x = k$: vertical line parallel to y -axis at $x = k$ (e.g. $x = 3$)
- ▶ $y = k$: horizontal line parallel to x -axis at $y = k$ (e.g. $y = -2$)
- ▶ $y = mx$: line passing through origin with slope m



How to Plot a Linear Equation

- Step 1: Put $x = 0 \rightarrow$ find y (y -intercept point)
- Step 2: Put $y = 0 \rightarrow$ find x (x -intercept point)
- Step 3: Choose one more x value $\rightarrow$ find y (third point for checking)
- Step 4: Plot the 3 points and join them with a straight line
- Step 5: Extend the line in both directions with arrows

⚠ Common Mistakes

- Any single equation in two variables does NOT have a unique solution — it has infinitely many.



- The line $x = 0$ is the y-axis. The line $y = 0$ is the x-axis.
- Always mark at least 3 points to be sure your graph is correct.

3.3 Worked Example

Worked Example

Q: Draw the graph of $2x + y = 6$ and find where it meets the axes.

Step 1: Put $x=0$: $0+y=6 \rightarrow y=6$. Point A=(0,6)

Step 2: Put $y=0$: $2x+0=6 \rightarrow x=3$. Point B=(3,0)

Step 3: Put $x=1$: $2+y=6 \rightarrow y=4$. Point C=(1,4)

Step 4: Plot A, B, C and join them. The line meets y-axis at (0,6) and x-axis at (3,0).

∴ Answer: y-intercept = 6, x-intercept = 3

3.4 Practice Questions

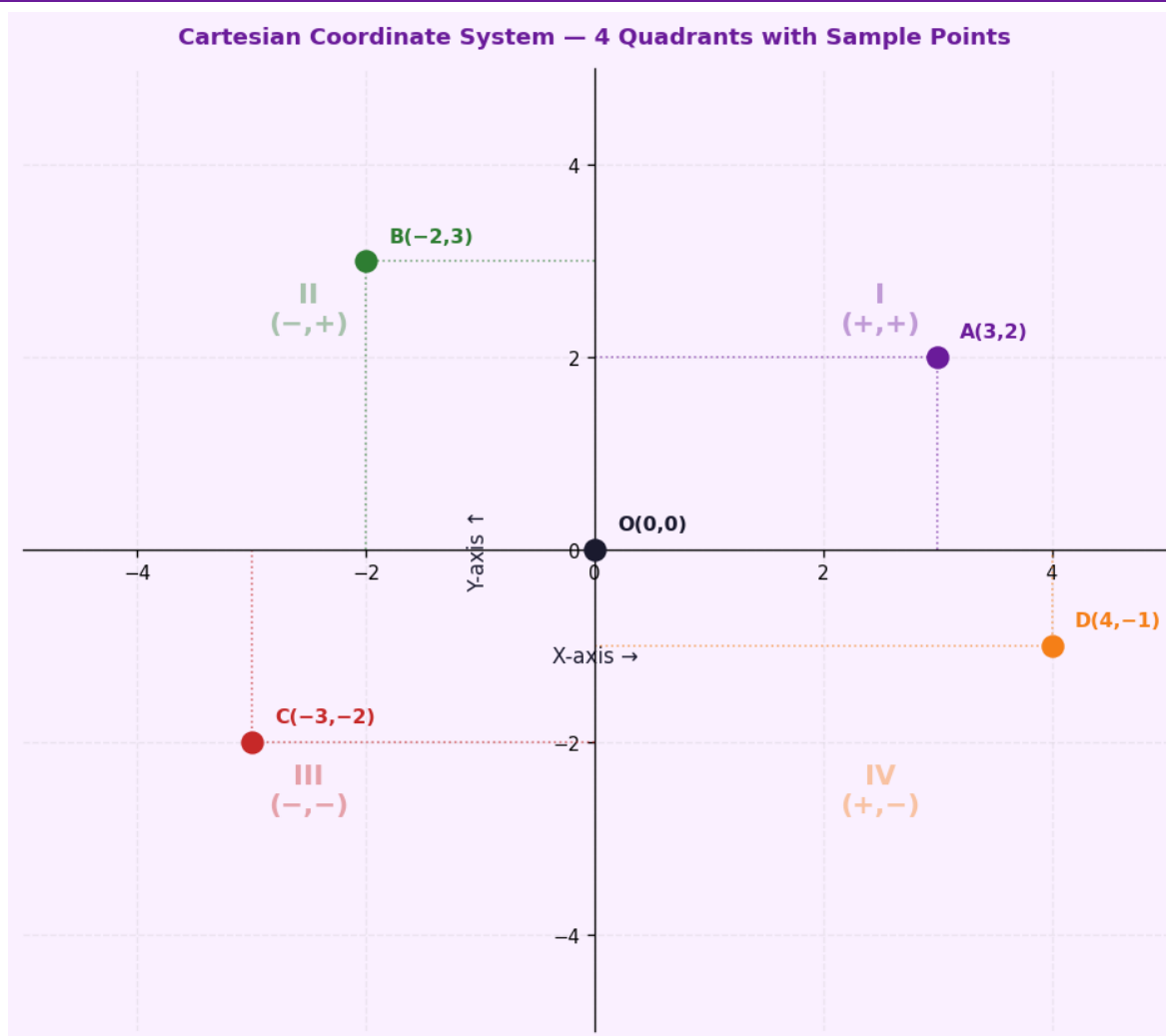
Year	SA / LA Question
Type	Write 4 solutions of $2x+y=7$. Are the solutions of $3x+5y=0$ real numbers?
Type	The cost of a notebook is twice that of a pen. Write a linear equation in two variables.
Type	Draw the graph of $x+y=7$. Shade the region between the line and the axes.
Type	If $x=2$, $y=1$ is a solution of $2x+3y=k$, find k .
Type	Write the equations of lines passing through the origin and through (2,5).

CHAPTER 4 · Unit III — 4 Marks

Coordinate Geometry

Cartesian Plane | Quadrants | Plotting Points

4.1 The Cartesian Plane



Graph 4.1: Cartesian plane showing 4 quadrants with sample points and dotted projection lines

Quadrant	x sign	y sign	Example
I (first)	Positive (+)	Positive (+)	(3, 5)
II (second)	Negative (-)	Positive (+)	(-4, 2)
III (third)	Negative (-)	Negative (-)	(-3, -6)
IV (fourth)	Positive (+)	Negative (-)	(5, -2)

► x-axis: horizontal axis ($y = 0$ on this axis)

- ▶ y-axis: vertical axis ($x = 0$ on this axis)
- ▶ Origin $O(0,0)$: intersection of the two axes
- ▶ Abscissa: x-coordinate of a point | Ordinate: y-coordinate of a point
- ▶ A point on x-axis has coordinates $(x, 0)$ | A point on y-axis has coordinates $(0, y)$

Special Points

- Axes points: $(a, 0)$ lies on x-axis; $(0, b)$ lies on y-axis
- Mirror of (x, y) in x-axis = $(x, -y)$ | Mirror in y-axis = $(-x, y)$
- Mirror of (x, y) in origin = $(-x, -y)$
- All points with equal x-coordinate lie on a vertical line
- All points with equal y-coordinate lie on a horizontal line

4.2 Worked Example

➔ Worked Example

Q: In which quadrant/axis do these points lie? $A(2, -3)$, $B(-5, 4)$, $C(0, -7)$, $D(-6, -2)$

Step 1: $A(2, -3)$: $x > 0$, $y < 0 \rightarrow$ Quadrant IV

Step 2: $B(-5, 4)$: $x < 0$, $y > 0 \rightarrow$ Quadrant II

Step 3: $C(0, -7)$: $x = 0 \rightarrow$ lies on the y-axis

Step 4: $D(-6, -2)$: $x < 0$, $y < 0 \rightarrow$ Quadrant III

∴ Answer: A: Q-IV, B: Q-II, C: y-axis, D: Q-III

4.3 Practice Questions

Year	SA / LA Question
Type	Plot the points $(1, 3), (2, -1), (-4, 0), (0, -3)$ on a Cartesian plane and state their quadrants.
Type	The point $(0, 4)$ lies on which axis? What is its distance from the origin?
Type	If a point $P(a, b)$ lies in the third quadrant, what are the signs of a and b ?
Type	Draw a triangle with vertices $A(1, 2)$, $B(4, 2)$, $C(3, 5)$. Find its base and height.
Type	The point $(-3, 0)$ lies on the x-axis. True or False? Justify.

CHAPTER 5 · Unit IV — Part of 27 Marks

Introduction to Euclid's Geometry

Axioms | Postulates | Theorems

5.1 Euclid's Approach

Euclid's Key Axioms & Postulates

Postulate 1 A straight line can be drawn from any point to any other point.

Postulate 2 A finite line can be extended indefinitely.

Postulate 3 A circle can be drawn with any centre and radius.

Postulate 4 All right angles are equal.

Postulate 5 Only ONE line parallel to a given line passes through a point not on it.

Axiom: Things equal to the same thing are equal to each other.

Lines and Angles

Alternate interior angles are EQUAL
Co-interior angles are SUPPLEMENTARY

Graph 5.1: Euclid's 5 postulates (left) and parallel lines with transversal angles (right)

- ▶ Euclid organised geometry using DEFINITIONS, AXIOMS, POSTULATES, and THEOREMS
- ▶ Axioms: self-evident truths accepted without proof (apply to all mathematics)
- ▶ Postulates: self-evident truths specific to geometry
- ▶ Theorem: a statement that can be PROVED from axioms and postulates

5.2 Euclid's Five Postulates

Postulate	Statement
P1	A straight line segment can be drawn joining any two points.
P2	Any straight line segment can be extended indefinitely in a straight line.
P3	Given any straight line segment, a circle can be drawn having the segment as radius and one endpoint as centre.
P4	All right angles are congruent (equal).
P5 (Parallel Postulate)	If a line segment intersects two straight lines forming two interior angles on the same side that sum to less than 180° , then the two lines, if extended indefinitely, meet on the side on which the angles sum to less than 180° .

Equivalent to Postulate 5 (Playfair's Axiom)

- Through a given point NOT on a line, exactly ONE line can be drawn parallel to the given line.
- This is the modern equivalent of Euclid's 5th postulate.
- Non-Euclidean geometries arise when this postulate is modified.

5.3 Key Axioms

Axiom	Statement
A1	Things equal to the same thing are equal to each other. ($a=b, b=c \Rightarrow a=c$)
A2	If equals are added to equals, the wholes are equal.
A3	If equals are subtracted from equals, the remainders are equal.
A4	Things which coincide with one another are equal to each other.
A5	The whole is greater than the part.

5.4 Practice Questions

Year	SA / LA Question
Type	State the axiom used: 'If $AB = PQ$ and $PQ = XY$, then $AB = XY$ '.
Type	If $AC = BD$, prove that $AB = CD$ where B is between A,C and C is between B,D.
Type	Prove: Two distinct lines cannot have more than one point in common.
Type	State Playfair's axiom. How is it related to Euclid's 5th Postulate?
Type	Give an example from daily life of each of Euclid's first 3 postulates.

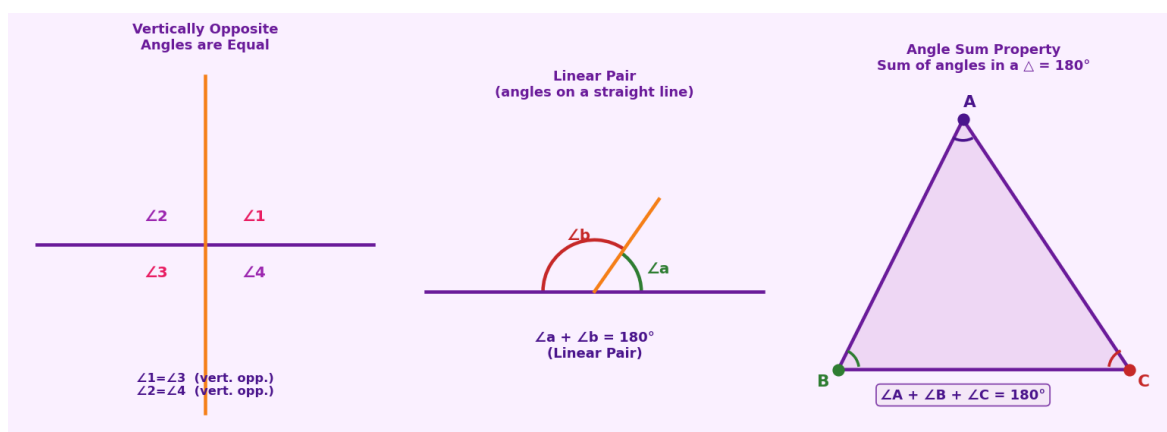
CHAPTER 6 · Unit IV — Part of 27 Marks

Lines and Angles

Angle Pairs | Parallel Lines | Transversal | Angle Sum

6.1 Types of Angles

Angle Type	Measure	Example
Acute	$> 0^\circ$ and $< 90^\circ$	$45^\circ, 60^\circ$
Right	$= 90^\circ$	Corner of a square
Obtuse	$> 90^\circ$ and $< 180^\circ$	$120^\circ, 150^\circ$
Straight	$= 180^\circ$	A straight line
Reflex	$> 180^\circ$ and $< 360^\circ$	270°
Complete	$= 360^\circ$	Full rotation



Graph 6.1: Vertically opposite angles (left), linear pair (centre), angle sum in triangle (right)

6.2 Angle Pair Relationships

Pair	Definition	Key Property
Complementary	Sum = 90°	30° and 60° are complementary
Supplementary	Sum = 180°	110° and 70° are supplementary
Linear Pair	Adjacent, supplementary, on a straight line	Always sum to 180°
Vertically Opposite	Formed when two lines intersect, opposite to each other	Always EQUAL

6.3 Key Theorems



Theorem 1: Linear Pair Axiom ★

- If a ray stands on a line, then the sum of two adjacent angles formed = 180°
- Converse: If sum of two adjacent angles = 180° , their non-common arms form a straight line



Theorem 2: Vertically Opposite Angles (PROVE required)

- Statement: If two lines intersect each other, vertically opposite angles are equal
- Proof: Lines AB and CD intersect at O. $\angle AOC + \angle BOC = 180^\circ$ (linear pair)
- $\angle BOC + \angle BOD = 180^\circ$ (linear pair). $\therefore \angle AOC = \angle BOD$. Similarly $\angle AOD = \angle BOC$.

6.4 Parallel Lines and Transversal



Angle Pairs with Parallel Lines $l_1 \parallel l_2$ (cut by transversal)

- Corresponding angles: EQUAL ($\angle 1 = \angle 5$, $\angle 2 = \angle 6$, $\angle 3 = \angle 7$, $\angle 4 = \angle 8$)
- Alternate interior angles: EQUAL ($\angle 3 = \angle 6$, $\angle 4 = \angle 5$)
- Alternate exterior angles: EQUAL ($\angle 1 = \angle 8$, $\angle 2 = \angle 7$)
- Co-interior (same-side interior) angles: SUPPLEMENTARY ($\angle 3 + \angle 5 = 180^\circ$)
- Converse: If corresponding angles are equal $\rightarrow$ lines are parallel

6.5 Angle Sum in a Triangle



Angle Sum Property of Triangle

- Statement: Sum of interior angles of a triangle = 180°
- Proof: Through A, draw line DE $\parallel$ BC. Then $\angle DAB = \angle ABC$ (alternate angles), $\angle EAC = \angle ACB$ (alternate angles).
- $\angle DAB + \angle BAC + \angle EAC = 180^\circ$ (straight line). $\therefore \angle ABC + \angle BAC + \angle ACB = 180^\circ$.

► Exterior angle of triangle = sum of two non-adjacent interior angles (exterior angle theorem)

⚠ Common Mistakes

- Linear pair: angles must be ADJACENT and supplementary (share a common arm and vertex).
- Vertically opposite angles: only when TWO straight lines intersect, not just any two angles.
- Alternate angles: only equal when lines are PARALLEL. Don't assume equality otherwise.
- Exterior angle: it equals the sum of remote interior angles, not just one of them.



6.6 Worked Example

Worked Example

Q: Two lines AB and CD intersect at O. If $\angle AOC = 50^\circ$, find all four angles.

Step 1: $\angle AOC = 50^\circ$ (given)

Step 2: $\angle BOD = 50^\circ$ (vertically opposite to $\angle AOC$)

Step 3: $\angle AOD = 180^\circ - 50^\circ = 130^\circ$ (linear pair with $\angle AOC$)

Step 4: $\angle BOC = 130^\circ$ (vertically opposite to $\angle AOD$)

∴ Answer: $\angle AOC = \angle BOD = 50^\circ$, $\angle AOD = \angle BOC = 130^\circ$

6.7 Practice Questions

Year	SA / LA Question
Type	If two parallel lines are cut by a transversal and one co-interior angle is 65° , find the other.
Type	The angles of a triangle are in ratio 3:4:5. Find each angle.
Type	Prove: If the sum of co-interior angles is 180° , then the lines are parallel.
Type	An exterior angle of a triangle is 120° . If interior opposite angles are equal, find them.
Type	Two lines are perpendicular to the same line. Prove they are parallel to each other.

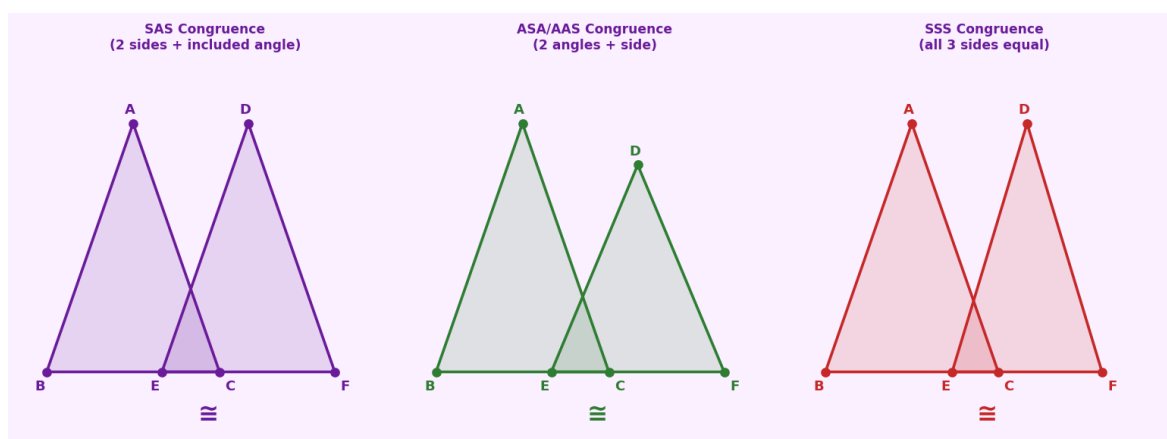
CHAPTER 7 · Unit IV — Part of 27 Marks

Triangles

Congruence Criteria | Properties | Inequalities

7.1 Congruence of Triangles

- ▶ Two triangles are CONGRUENT ($\cong$) if they have exactly the same shape and size
- ▶ Corresponding parts of congruent triangles are equal (CPCT / CPCTC)



Graph 7.1: SAS (left), ASA/AAS (centre), SSS (right) congruence criteria

Criterion	Full Name	Conditions	Note
SAS	Side-Angle-Side	2 sides equal + included angle equal	Angle MUST be between the 2 sides
ASA	Angle-Side-Angle	2 angles equal + included side equal	Side between the 2 angles
AAS	Angle-Angle-Side	2 angles equal + non-included side equal	Side not between the angles
SSS	Side-Side-Side	All 3 sides equal	—
RHS	Right-Hyp-Side	Right angle + hypotenuse + one side equal	Only for right triangles

CPCT Rule

- If $\triangle ABC \cong \triangle DEF$, then: $AB=DE$, $BC=EF$, $CA=FD$ (sides) AND $\angle A=\angle D$, $\angle B=\angle E$, $\angle C=\angle F$ (angles)
- This is used AFTER proving congruence to conclude equal sides or angles.
- Order of vertices matters: $\triangle ABC \cong \triangle DEF$ means $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$

7.2 Key Theorems about Triangles

Theorem: Angles Opposite Equal Sides

- If two sides of a triangle are equal, then angles opposite to them are equal.
- In $\triangle ABC$: $AB = AC \Rightarrow \angle B = \angle C$ (isosceles triangle property)
- Converse: If $\angle B = \angle C$, then $AB = AC$

Theorem: Side Opposite Greater Angle

- In a triangle, the side opposite the greater angle is greater.
- If $\angle A > \angle B$, then $BC > AC$ (the side opposite $\angle A$ is BC)
- Converse: The angle opposite the greater side is greater.

Triangle Inequality

- Sum of any two sides of a triangle $>$ third side
- $AB + BC > CA$ | $BC + CA > AB$ | $CA + AB > BC$
- This means: to form a triangle, any two sides must together be longer than the third

Common Mistakes

- SSA is NOT a valid congruence criterion (two different triangles can have same SS and A).
- AAA is NOT a congruence criterion (only proves similarity, not congruence).
- CPCT can only be used AFTER you have proved the triangles are congruent.
- In RHS: the right angle is at the vertex, hypotenuse is the longest side (opposite 90°).

7.3 Worked Example

Worked Example

Q: In $\triangle ABC$ and $\triangle PQR$, $AB=PQ$, $\angle B=\angle Q$, $BC=QR$. Prove $\triangle ABC \cong \triangle PQR$.

Step 1: In $\triangle ABC$ and $\triangle PQR$:

Step 2: $AB = PQ$ (given)

Step 3: $\angle B = \angle Q$ (given)

Step 4: $BC = QR$ (given)

Step 5: $\therefore \triangle ABC \cong \triangle PQR$ (by SAS congruence criterion)

$\therefore$ Answer: $\triangle ABC \cong \triangle PQR$ by SAS

7.4 Practice Questions

Year	SA / LA Question
Type	In an isosceles $\triangle ABC$ where $AB=AC$, D is midpoint of BC. Prove $\triangle ABD \cong \triangle ACD$.

Type	If the altitudes of two sides of a triangle are equal, prove the triangle is isosceles.
Type	In $\triangle ABC$, $\angle B = \angle C$. Prove $AB = AC$.
Type	Show that in a right triangle, the hypotenuse is the longest side.
Type	Prove: Sum of any two sides of a triangle $>$ the third side (triangle inequality).

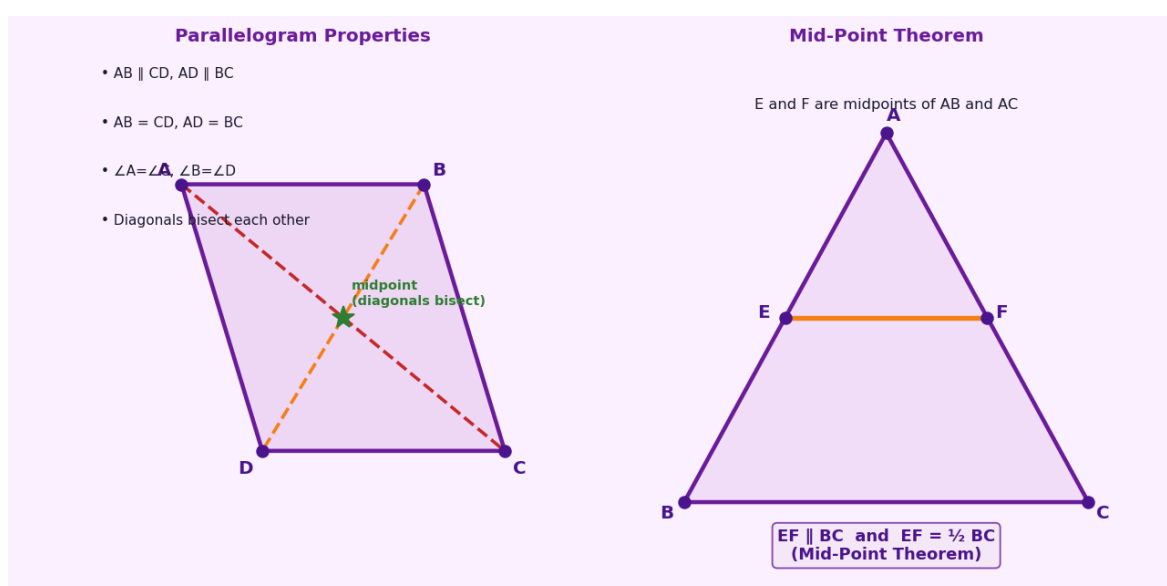
CHAPTER 8 · Unit IV — Part of 27 Marks

Quadrilaterals

Parallelogram Properties | Mid-Point Theorem

8.1 Types of Quadrilaterals

Type	Properties
Parallelogram	Opposite sides $\parallel$ and equal Opposite angles equal Diagonals bisect each other
Rectangle	Parallelogram + all angles = 90° Diagonals equal in length
Rhombus	Parallelogram + all sides equal Diagonals bisect at right angles
Square	Rectangle + Rhombus All sides equal + all angles 90° Diagonals equal + bisect at 90°
Trapezium	One pair of opposite sides parallel (called parallel sides)
Kite	Two pairs of consecutive sides equal One diagonal bisects the other at 90°



Graph 8.1: Parallelogram with properties (left) and Mid-Point Theorem in triangle (right)

8.2 Properties of Parallelogram (Theorems)



Key Parallelogram Theorems

1. A diagonal of a parallelogram divides it into two congruent triangles. (PROVE required)
2. In a parallelogram: opposite sides are equal (and converse).
3. In a parallelogram: opposite angles are equal (and converse).
4. In a parallelogram: diagonals bisect each other (and converse).

- 5. A quadrilateral is a parallelogram if a pair of opposite sides is BOTH parallel AND equal.

Proving Diagonal Divides Parallelogram (Theorem 1)

- In $\parallel\text{gm}$ ABCD, diagonal AC divides it into $\triangle ABC$ and $\triangle CDA$
- $AB \parallel DC \Rightarrow \angle BAC = \angle DCA$ (alternate interior angles)
- $AD \parallel BC \Rightarrow \angle BCA = \angle DAC$ (alternate interior angles)
- AC is common. So by ASA: $\triangle ABC \cong \triangle CDA \therefore AB=CD, AD=BC$ (CPCT)

8.3 Mid-Point Theorem ★

Mid-Point Theorem (PROVE required)

- Statement: The line segment joining the midpoints of two sides of a triangle is:
- (i) Parallel to the third side, AND (ii) Half the length of the third side
- If E and F are midpoints of AB and AC in $\triangle ABC$, then $EF \parallel BC$ and $EF = \frac{1}{2} BC$
- Converse: Line through midpoint of one side, parallel to another side bisects the third

Common Mistakes

- Angle sum of quadrilateral = 360° (NOT 180° , that's for triangles).
- Diagonals of parallelogram are NOT necessarily equal (they're equal only in rectangle).
- Diagonals of parallelogram bisect EACH OTHER but NOT the angles (rhombus does that).
- Mid-point theorem: BOTH conditions must hold (parallel AND half) — don't state just one.

8.4 Worked Example

Worked Example

Q: In parallelogram ABCD, $\angle A = 70^\circ$. Find all angles.

Step 1: $\angle C = 70^\circ$ (opposite angles of parallelogram are equal)

Step 2: $\angle A + \angle B = 180^\circ$ (co-interior angles between parallel sides AB and DC)

Step 3: $70^\circ + \angle B = 180^\circ \Rightarrow \angle B = 110^\circ$

Step 4: $\angle D = 110^\circ$ (opposite to $\angle B$)

$\therefore$ Answer: $\angle A = \angle C = 70^\circ, \angle B = \angle D = 110^\circ$

8.5 Practice Questions

Year	SA / LA Question
Type	ABCD is a parallelogram. If $\angle A = 3x$, $\angle B = 2x + 20^\circ$, find x and all angles.
Type	Prove that the diagonals of a rhombus bisect each other at right angles.

Type	E,F are midpoints of AB,CD of $\parallel$ gm ABCD. Show that AECF is a $\parallel$ gm.
Type	Prove the Mid-Point Theorem. Use it to find EF if BC=12 cm.
Type	In a square ABCD, diagonals AC and BD meet at O. Show $\triangle AOB$ is isosceles.

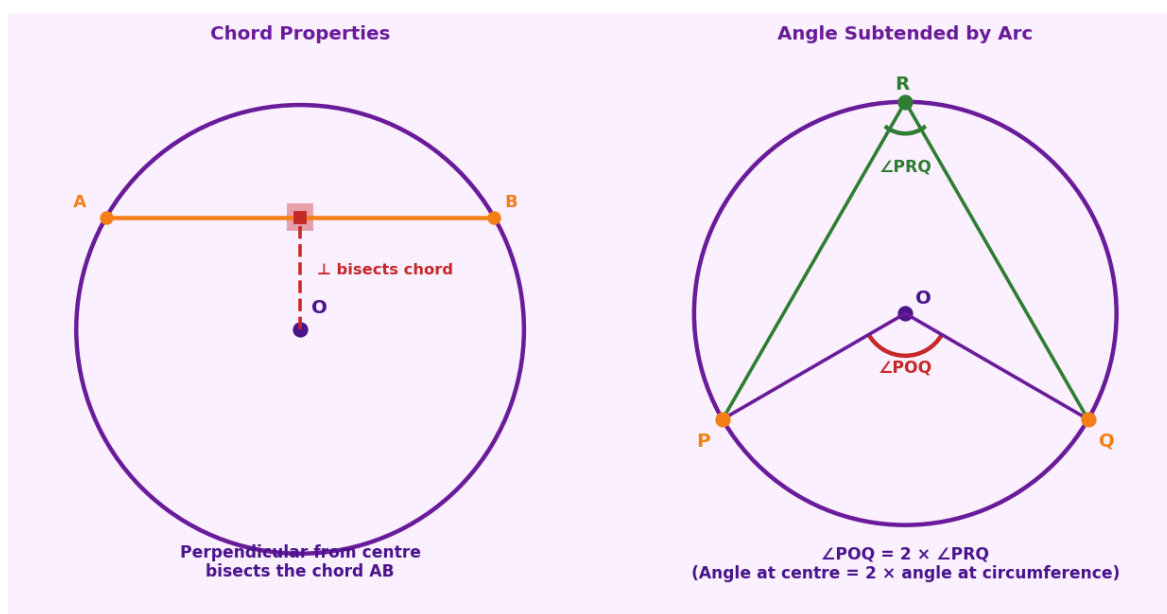
CHAPTER 9 · Unit IV — Part of 27 Marks

Circles

Chord Properties | Angle Theorems | Cyclic Quadrilateral

9.1 Key Terms and Chord Theorems

Term	Definition
Chord	Line segment with both endpoints on the circle
Diameter	Longest chord passing through centre = $2r$
Arc	Part of the circle between two points
Sector	Region bounded by two radii and an arc
Segment	Region between a chord and the arc it cuts off
Secant	A line that intersects the circle at two points
Tangent	A line that touches the circle at exactly one point

Graph 9.1: Chord properties — perpendicular bisects chord (left) | angle at centre = $2 \times$ angle at circumference (right)

9.2 Chord Theorems



Equal Chords Theorem

- Equal chords of a circle (or equal circles) subtend equal angles at the centre
- Converse: If chords subtend equal angles at centre, they are equal



- Equal chords are equidistant from the centre (and converse)



Perpendicular from Centre Theorem

- The perpendicular from the centre of a circle to a chord BISECTS the chord
- Converse: The line drawn through the centre to bisect a chord is perpendicular to it
- Proof: Drop perpendicular OM to chord AB. Show $\triangle OMA \cong \triangle OMB$ by RHS $\Rightarrow AM = MB$

9.3 Angle Theorems (All PROVE required)



Theorem: Angle at Centre = $2 \times$ Angle at Circumference ★

- The angle subtended by an arc at the centre is double the angle at any point on the remaining arc
- $\angle AOB = 2 \times \angle ACB$ where O = centre, C is on the major arc
- Proof: Join CO extended to D. Use exterior angle of triangle twice.



Theorem: Angles in the Same Segment are Equal

- Angles subtended by the same arc at any point on the remaining part of the circle are equal
- $\angle ACB = \angle ADB$ when C and D are on the same arc
- This follows from: both = half of $\angle AOB$ (angle at centre)



Theorem: Cyclic Quadrilateral ★

- The sum of opposite angles of a cyclic quadrilateral (all vertices on a circle) = 180°
- $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$ in cyclic quadrilateral ABCD
- Converse: If opposite angles of a quadrilateral sum to 180° , it is cyclic

- Angle in a semicircle = 90° (angle in a semicircle is a right angle)
- Angles in same segment are EQUAL | Angle at centre = $2 \times$ angle at circumference

⚠ Common Mistakes

- Angle at centre is DOUBLE the angle at circumference, not the other way.
- Equal chords theorem: chords must be in the SAME circle or equal circles.
- Cyclic quadrilateral: OPPOSITE angles sum to 180° , not adjacent angles.
- Angle in semicircle = 90° only when the angle is inscribed in a semicircle.

9.4 Worked Example

➡ Worked Example



Q: O is centre of circle. $\angle AOB = 100^\circ$. Find $\angle ACB$ where C is on the major arc.

Step 1: $\angle AOB = 100^\circ$ (angle at centre)

Step 2: $\angle ACB = \frac{1}{2} \times \angle AOB$ (angle at circumference = half angle at centre)

Step 3: $\angle ACB = \frac{1}{2} \times 100^\circ = 50^\circ$

$\therefore$ Answer: $\angle ACB = 50^\circ$

9.5 Practice Questions

Year	SA / LA Question
Type	A chord AB = 6 cm. Centre O is 4 cm from A. Find the distance from O to chord AB.
Type	Prove: The perpendicular from centre to a chord bisects the chord.
Type	In a circle with centre O, $\angle POQ = 130^\circ$. Find $\angle PAQ$ where A is on major arc.
Type	Prove that angles in the same segment of a circle are equal.
Type	ABCD is a cyclic quadrilateral with $\angle A = 80^\circ$, $\angle B = 70^\circ$. Find $\angle C$ and $\angle D$.

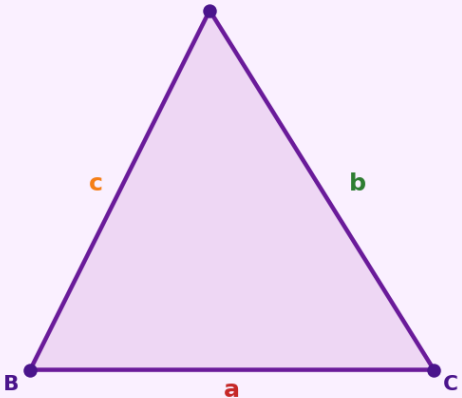
CHAPTER 10 · Unit V — Part of 13 Marks

Heron's Formula

Area of Triangle | Scalene | Quadrilateral application

10.1 Heron's Formula

Heron's Formula



$s = (a+b+c)/2$ (semi-perimeter)

Heron's Formula — Example

Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$

where $s = (a+b+c)/2$ (semi-perimeter)

Example: $a=5, b=6, c=7$

$s = (5+6+7)/2 = 9$

$s-a = 4, s-b = 3, s-c = 2$

Area = $\sqrt{(9 \times 4 \times 3 \times 2)} = \sqrt{216}$

$= 6\sqrt{6} \approx 14.7 \text{ cm}^2$

Graph 10.1: Triangle with labelled sides (left) and worked example using Heron's formula (right)

Heron's Formula ★

- Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$
- where $s = (a+b+c)/2$ is the SEMI-PERIMETER
- a, b, c are the three sides of the triangle
- Works for ANY triangle — scalene, isosceles, equilateral, right-angled

► Area = $\sqrt{s(s-a)(s-b)(s-c)}$

► Semi-perimeter $s = \frac{a+b+c}{2}$

10.2 Special Cases

Triangle Type	Simplified Area Formula	When to use
Equilateral (all sides = a)	$(\sqrt{3}/4)a^2$	All three sides equal
Right-angled (base b, height h)	$(1/2) \times b \times h$	Right angle known
Isosceles (equal sides = a, base = b)	$(b/4)\sqrt{(4a^2-b^2)}$	Two sides equal

10.3 Area of Quadrilateral using Heron's Formula



Method: Divide quadrilateral into two triangles

- Step 1: Draw a diagonal to divide the quadrilateral into two triangles
- Step 2: Apply Heron's formula to each triangle separately
- Step 3: Total area = sum of areas of both triangles
- This works for any quadrilateral where you know all 4 sides and one diagonal

⚠ Common Mistakes

- Semi-perimeter $s = (a+b+c)/2$, NOT $(a+b+c)$. Don't forget to divide by 2.
- Inside the square root: $s(s-a)(s-b)(s-c)$ — must compute all four factors.
- Equilateral triangle: use $(\sqrt{3}/4)a^2$ only when ALL THREE sides are equal.
- Always verify: result should be positive and make geometric sense.

10.4 Worked Example

➡ Worked Example

Q: Find the area of a triangle with sides 7 cm, 8 cm, 9 cm.

Step 1: $s = (7+8+9)/2 = 24/2 = 12$ cm

Step 2: $s-a = 12-7 = 5$ | $s-b = 12-8 = 4$ | $s-c = 12-9 = 3$

Step 3: Area = $\sqrt{[12 \times 5 \times 4 \times 3]} = \sqrt{720} = \sqrt{(144 \times 5)} = 12\sqrt{5}$

∴ Answer: Area = $12\sqrt{5} \approx 26.83$ cm²

10.5 Practice Questions

Year	SA / LA Question
Type	Find area of triangle with sides 5,6,7 cm using Heron's formula.
Type	Triangular park has sides 40m, 32m, 24m. Find cost of planting grass at ₹50 per m ² .
Type	Rhombus with side 6cm and diagonal 8cm. Find its area using Heron's formula.
Type	Quadrilateral ABCD: AB=3, BC=4, CD=4, DA=5, diagonal AC=5. Find area.
Type	An isosceles triangle has perimeter 30cm and equal sides 12cm each. Find area.

CHAPTER 11 · Unit V — Part of 13 Marks

Surface Areas and Volumes

Cylinder | Cone | Sphere | Hemisphere

11.1 Complete Formula Table

Surface Areas and Volumes — Class 9 Shapes (r=3 cm, h/l as given)		
Cylinder $r=3, h=7$ CSA = $2\pi rh$ $= 2\pi \times 3 \times 7 = 42\pi$ TSA = $2\pi r(r+h)$ $= 2\pi \times 3 \times 10 = 60\pi$ V = $\pi r^2 h$ $= \pi \times 9 \times 7 = 63\pi$	Cone $r=3, l=5, h=4$ CSA = πrl $= \pi \times 3 \times 5 = 15\pi$ TSA = $\pi r(r+l)$ $= \pi \times 3 \times 8 = 24\pi$ V = $\frac{1}{3}\pi r^2 h$ $= \frac{1}{3} \times \pi \times 9 \times 4 = 12\pi$	Sphere & Hemisphere $r=3$ Sphere SA = $4\pi r^2$ $= 36\pi$ Sphere V = $\frac{4}{3}\pi r^3$ $= 36\pi$ Hemisphere CSA = $2\pi r^2$ $= 18\pi$ Hemisphere V = $\frac{2}{3}\pi r^3$ $= 18\pi$

Graph 11.1: Formula summary cards for cylinder, cone, and sphere/hemisphere

Shape	Curved SA	Total SA	Volume	Slant / Other
Cube (side a)	$4a^2$	$6a^2$	a^3	Diagonal = $a\sqrt{3}$
Cuboid (l,b,h)	$2h(l+b)$	$2(lb+bh+hl)$	lbh	Diagonal = $\sqrt{l^2+b^2+h^2}$
Cylinder (r,h)	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$	l (axis) = h
Cone (r,h,l)	πrl	$\pi r(r+l)$	$\frac{1}{3}\pi r^2 h$	$l = \sqrt{h^2+r^2}$
Sphere (r)	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$	—
Hemisphere (r)	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$	Flat face = πr^2

Key derived formulas:

- **Cone slant height:** $l = \sqrt{h^2 + r^2}$
- **Cuboid diagonal:** $d = \sqrt{l^2 + b^2 + h^2}$

11.2 Important Conversions

💡 Unit Conversions for Volume Problems

- $1 \text{ m}^3 = 1000 \text{ litres}$ (very commonly tested!)
- $1 \text{ litre} = 1000 \text{ cm}^3 = 1000 \text{ mL}$

- $1 \text{ cm}^3 = 1 \text{ mL}$
- $1 \text{ m} = 100 \text{ cm} \Rightarrow 1 \text{ m}^2 = 10000 \text{ cm}^2 \Rightarrow 1 \text{ m}^3 = 10^6 \text{ cm}^3$

⚠ Common Mistakes

- Cylinder TSA = $2\pi r(r+h)$. Don't forget the TWO circular ends.
- Cone TSA = $\pi r(r+l)$. Only ONE circular base (unlike cylinder).
- Hemisphere TSA = $3\pi r^2 = 2\pi r^2$ (curved) + πr^2 (flat base). Don't use $4\pi r^2$.
- Volume of cone = $\frac{1}{3} \times$ Volume of cylinder (same r and h). Factor of $\frac{1}{3}$ is critical.

11.3 Worked Example

➡ Worked Example

Q: A cone has radius 7 cm and slant height 25 cm. Find: CSA, TSA, Volume. ($\pi=22/7$)

Step 1: Find h: $l^2 = h^2 + r^2 \Rightarrow 25^2 = h^2 + 7^2 \Rightarrow h^2 = 625 - 49 = 576 \Rightarrow h = 24 \text{ cm}$

Step 2: CSA = $\pi r l = (22/7) \times 7 \times 25 = 550 \text{ cm}^2$

Step 3: TSA = $\pi r(r+l) = (22/7) \times 7 \times (7+25) = 22 \times 32 = 704 \text{ cm}^2$

Step 4: Volume = $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times (22/7) \times 49 \times 24 = 1232 \text{ cm}^3$

∴ Answer: CSA=550 cm², TSA=704 cm², V=1232 cm³

11.4 Practice Questions

Year	SA / LA Question
Type	A sphere has surface area 154 cm^2 . Find its radius and volume. ($\pi=22/7$)
Type	A cylindrical tank of radius 1.5m and height 7m. Find capacity in litres.
Type	Cone and cylinder have same base radius and height. Compare their volumes.
Type	A solid hemisphere is placed on a cylinder. Both have radius 7cm, height of cylinder = 12cm. Find TSA.
Type	Curved surface area of sphere = CSA of cylinder (same radius). Find ratio of their volumes.

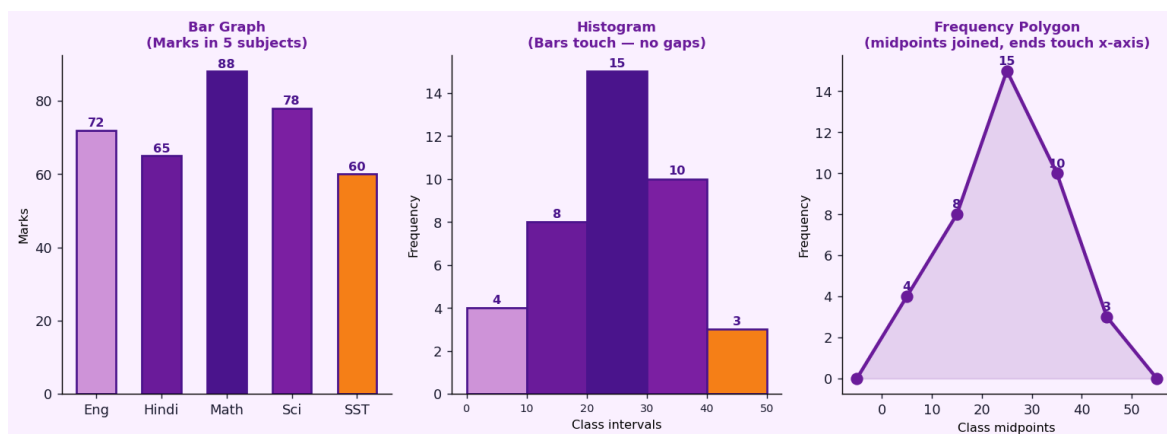
CHAPTER 12 · Unit VI — 6 Marks

Statistics

Data Collection | Bar Graph | Histogram | Frequency Polygon

12.1 Types of Data and Frequency Distribution

Term	Definition	Example
Raw data	Data collected in its original form, not organised	Heights: 142,165,150,158...
Grouped data	Data arranged in class intervals	0–10, 10–20, 20–30...
Frequency	Number of times a value or class occurs	Class 10–20 has frequency 8
Class width	Difference between upper and lower boundaries	10–20: width = 10
Class mark	Midpoint of a class interval	Midpoint of 10–20 = 15
Frequency density	Frequency ÷ class width (for unequal widths)	$8 \div 10 = 0.8$



Graph 12.1: Bar graph (left), Histogram with touching bars (centre), Frequency Polygon (right)

12.2 Types of Graphical Representation

Graph	Used for	Key Feature	How to draw
Bar graph	Discrete/categorical data	Bars separated by gaps	Equal width bars, height = frequency
Histogram	Continuous grouped data	Bars touch (NO gaps)	Width = class interval, height = frequency

Frequency polygon	Continuous data (trend)	Line graph from midpoints	Join class marks with frequency, close at x-axis
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Histogram vs Bar Graph

- Bar Graph: gaps between bars | Histogram: NO gaps (bars touch each other)
- Bar Graph: x-axis has categories | Histogram: x-axis has continuous class intervals
- Frequency Polygon: join midpoints of tops of histogram bars; close at x-axis
- For unequal class widths in histogram: use frequency DENSITY (not frequency) on y-axis

Drawing a Frequency Polygon

- Step 1: Draw the histogram for the frequency distribution
- Step 2: Mark the midpoint (class mark) at the top of each bar
- Step 3: Add two fictitious classes (one before and one after) with frequency 0
- Step 4: Join all midpoints (including the zero-frequency endpoints) with straight lines
- The resulting polygon touches the x-axis at both ends

Common Mistakes

- Histogram bars must TOUCH each other (no gaps) unlike bar graphs.
- Class mark = (lower boundary + upper boundary) / 2. Not the same as frequency.
- For frequency polygon: extend to x-axis at both ends (fictitious classes with f=0).
- y-axis in histogram = frequency (or frequency density if unequal class widths).

12.3 Measures of Central Tendency (Formative only)

Mean, Median, Mode of Ungrouped Data:

- ▶ **Mean** $\bar{x} = \frac{\sum x_i}{n}$ *Sum of all observations ÷ number of observations*
- ▶ **Median**: arrange in order; middle value (odd n) or average of two middle values (even n)
- ▶ **Mode**: the value that appears most frequently in the data

12.4 Worked Example

Worked Example

Q: The weights (kg) of 10 students: 42, 45, 48, 45, 50, 52, 45, 48, 52, 50. Find mean, median, mode.

Step 1: Arrange: 42, 45, 45, 45, 48, 48, 50, 50, 52, 52

Step 2: Mean = $(42+45+45+45+48+48+50+50+52+52)/10 = 477/10 = 47.7$ kg

Step 3: Median = average of 5th and 6th term = $(48+48)/2 = 48$ kg

Step 4: Mode = 45 kg (appears 3 times, the most frequent)

∴ Answer: Mean = 47.7 kg, Median = 48 kg, Mode = 45 kg

12.5 Practice Questions

Year	SA / LA Question
Type	Draw a histogram for: 10-20(5), 20-30(8), 30-40(15), 40-50(10), 50-60(7)
Type	Draw a frequency polygon for the data above.
Type	Ages of 20 students: 14,15,16,14,15,15,16,14,15,16,14,15,14,15,16,15,14,16,15,14. Find mean, median, mode.
Type	What is the difference between a bar graph and a histogram? Give examples.
Type	The mean of 5 numbers is 18. If one number is removed, the mean becomes 16. Find the removed number.

Master Quick Revision Sheet — All 12 Chapters (2025-26)

Chapter	★ Must-Know Formulas & Key Points
Ch 1 Real Numbers	$N \subset W \subset Z \subset Q \subset R$ Irr=non-term non-recur Conjugate rationalise $a^{(m/n)} = (\sqrt[n]{a})^m$
Ch 2 Polynomials	Rem Thm: $\text{rem} = p(a)$ Factor Thm: $p(a)=0 \Rightarrow (x-a)$ factor $(a+b)^2 = a^2 + 2ab + b^2$ $a^3 + b^3 + c^3 = 3abc$ if $a+b+c=0$
Ch 3 Linear Eqns	$ax+by+c=0$ ∞ solutions graph=straight line $x=k$: vertical, $y=k$: horizontal
Ch 4 Coordinate Geom	4 quadrants: I(+,+), II(-,+), III(-,-), IV(+,-) Origin=(0,0) Abscissa=x, Ordinate=y
Ch 5 Euclid	5 postulates Axioms: things equal to same thing are equal Playfair's axiom = Postulate 5
Ch 6 Lines & Angles	Vertically opp. = equal Linear pair = 180° Alt. int. angles = equal (parallel) Angle sum $\triangle = 180^\circ$
Ch 7 Triangles	SAS/ASA/AAS/SSS/RHS CPCT after congruence Isosceles: equal sides $\leftrightarrow$ equal angles
Ch 8 Quadrilaterals	Opp sides = equal Opp angles = equal Diagonals bisect Mid-pt: $EF \parallel BC$, $EF = \frac{1}{2}BC$
Ch 9 Circles	Perp from centre bisects chord Angle@centre = $2 \times$ angle@circumference Cyclic quad: opp angles = 180°
Ch 10 Heron's Formula	$s = (a+b+c)/2$ Area = $\sqrt{s(s-a)(s-b)(s-c)}$ Equil = $(\sqrt{3}/4)a^2$
Ch 11 SA & Volumes	Cone: $CSA = \pi rl$, $V = \frac{1}{3}\pi r^2 h$, $l = \sqrt{h^2 + r^2}$ Sphere: $SA = 4\pi r^2$, $V = \frac{4}{3}\pi r^3$ Hemi: $CSA = 2\pi r^2$
Ch 12 Statistics	Bar:gaps Histogram:no gaps Freq poly:join midpoints Mean = $\Sigma x/n$ Mode=most frequent

SA Exam Strategy 2025-26 (80 Marks Theory + 20 Internal)

Strategy	Detail
Paper Pattern	10 MCQ (1M) + 5 Very Short (2M) + 6 Short (3M) + 4 Long (5M) — check school pattern
High Yield Units	Geometry (27M) → Algebra (20M) → Mensuration (13M) → Number System (10M)
Proof Questions	State theorem → Given → To Prove → Construction (if needed) → Proof
Construction	Use compass and ruler; label all points; show all arcs

Identities	Memorise all 9 identities in Ch2 — they appear in almost every paper
Theorems	Theorem 1 (diagonal of gm), Angle at centre, Equal tangents — know proof by heart
Mensuration	Write the formula first, then substitute. Keep $\pi=22/7$ unless stated otherwise
Statistics	Practice drawing histogram and frequency polygon to scale. Bars must touch in histogram
Internal 20M	5M pen-paper test + 5M multiple assessment + 5M portfolio + 5M lab practical

Final Words

- This book covers 100% of the official CBSE 2025-26 Class 9 Maths syllabus.
- Geometry (Unit IV) carries 27 marks — master all theorems and their proofs!
- Visit sjmaths.com for free notes, video lectures and past paper solutions.
- Understand concepts first, then memorise formulas — not the other way around.
- All the best for your Class 9 exams! 🍀